

# Mamba learning rate and inner width

Exact original GraphCast loss | 1 degree | batch 4 | di16 and di32 | 10-day forecasts

## Completed slower, joint and faster Mamba comparisons

| Inner width | Mamba LR            | Final rollout | SWA rollout | SWA day 10 | SWA vs joint |
|-------------|---------------------|---------------|-------------|------------|--------------|
| di16        | Slower Mamba (0.3x) | 13.63%        | 14.70%      | 21.98%     | -0.97 pp     |
| di16        | Joint LR (1x)       | 15.26%        | 15.67%      | 22.44%     | +0.00 pp     |
| di16        | Faster Mamba (3x)   | 15.26%        | 16.34%      | 23.68%     | +0.67 pp     |
| di32        | Slower Mamba (0.3x) | 15.21%        | 15.82%      | 22.96%     | -0.48 pp     |
| di32        | Joint LR (1x)       | 15.69%        | 16.29%      | 23.32%     | +0.00 pp     |
| di32        | Faster Mamba (3x)   | 15.14%        | 15.73%      | 22.75%     | -0.56 pp     |

di16: slower Mamba 14.70% vs joint 15.67% rollout reduction (-0.97 percentage points).

di32: slower Mamba 15.82% vs joint 16.29% rollout reduction (-0.48 percentage points).

Increasing di16 to di32: joint +0.62 pp; slower Mamba +1.11 pp (SWA).

Different warm-up weights and one training seed limit causal conclusions.

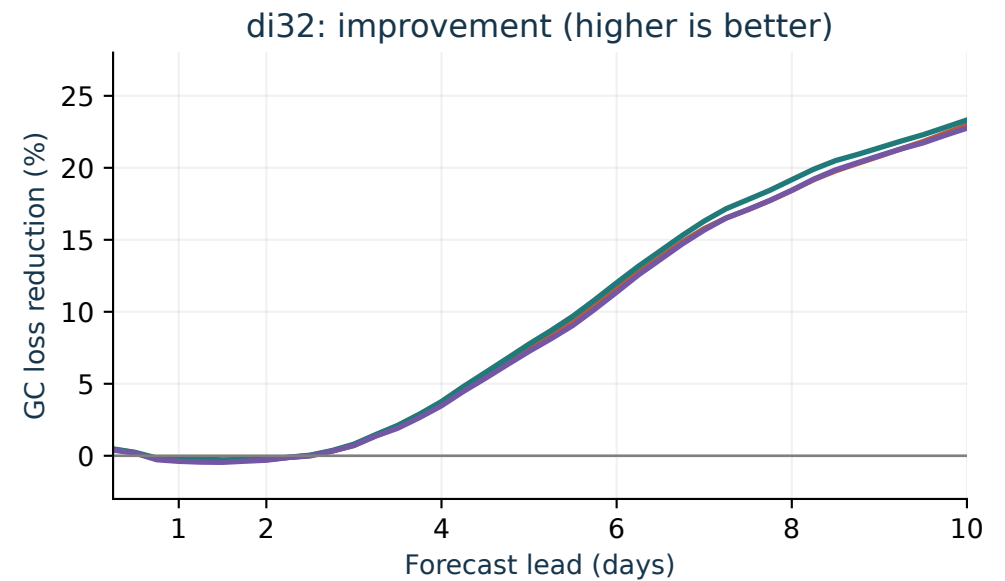
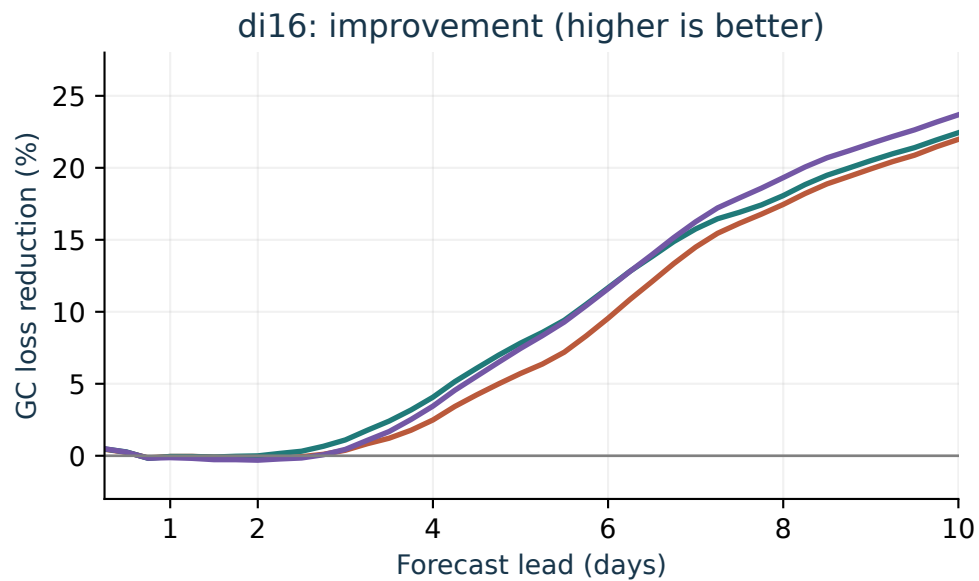
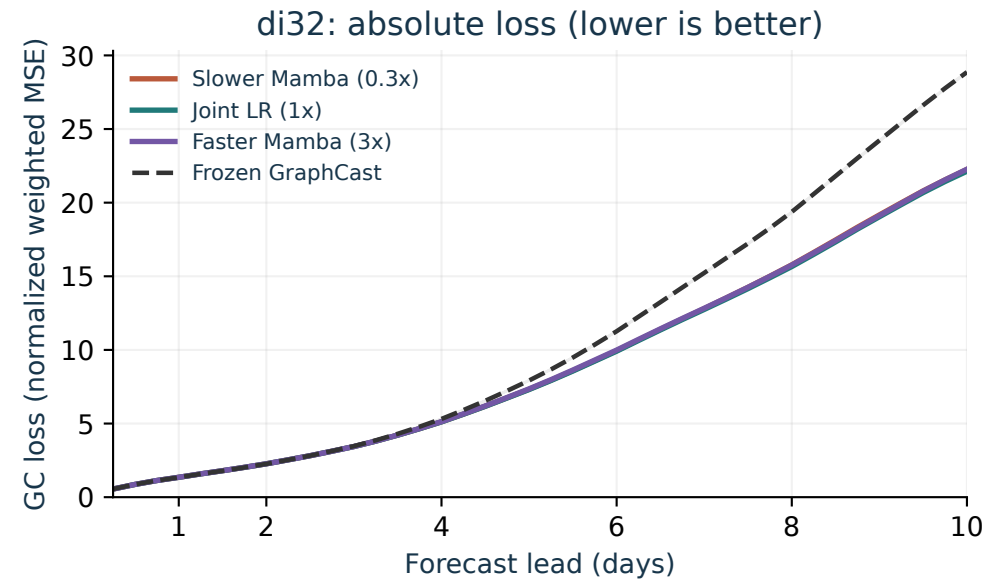
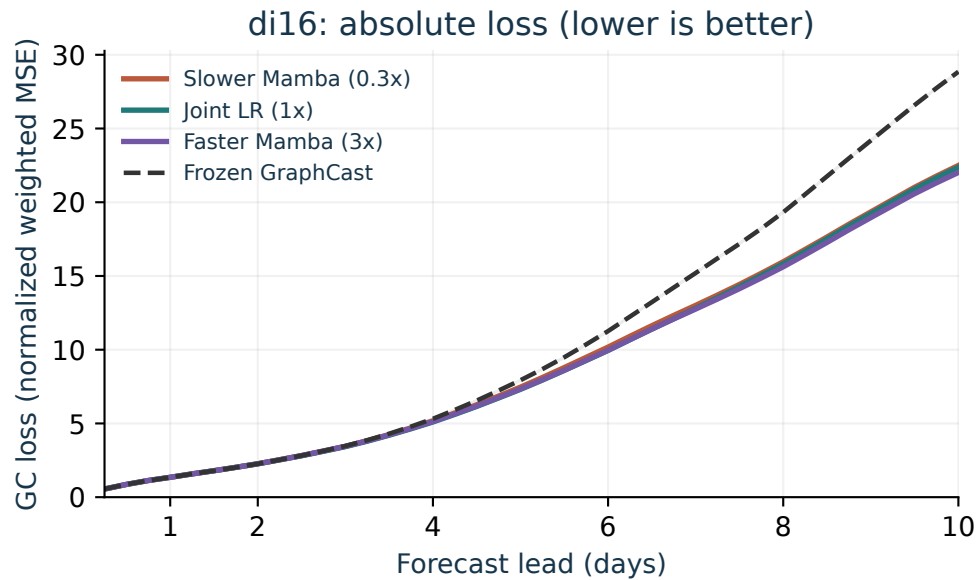
Higher reduction is better. All headline scores require 32 completed forecast starts.

Rollout: aggregate all 40 six-hour losses before taking the ratio. Day 10: the final lead only.

Final = branch update 2,000. SWA = average of branch checkpoints 500, 1,000, 1,500, 2,000.

# Forecast loss across all 10 days

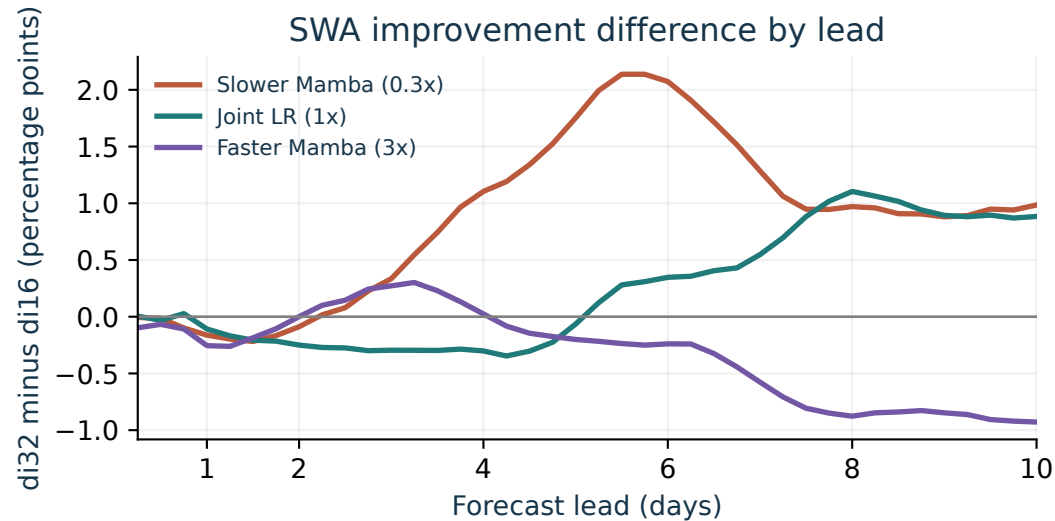
SWA checkpoints | complete 32-start evaluations | solid colors identify Mamba learning-rate multipliers



Each improvement uses its own matched baseline. Gray bands show baseline variation across completed runs.

# How the result depends on di

di is the Mamba inner dimension; spatial width stays 512 and BC groups stay 1



| SWA        | Day 1  | Day 5 | Day 10 |
|------------|--------|-------|--------|
| di16 slow  | -0.05% | 5.72% | 21.98% |
| di16 joint | -0.05% | 7.82% | 22.44% |
| di16 fast  | -0.13% | 7.45% | 23.68% |
| di32 slow  | -0.21% | 7.47% | 22.96% |
| di32 joint | -0.16% | 7.76% | 23.32% |
| di32 fast  | -0.39% | 7.25% | 22.75% |

| Arm             | Own warm-up | SWA rollout | Gain over own warm-up |
|-----------------|-------------|-------------|-----------------------|
| di16_joint      | 9.06%       | 15.67%      | +6.61 pp              |
| di16_slow_mamba | 7.13%       | 14.70%      | +7.58 pp              |
| di32_joint      | 7.37%       | 16.29%      | +8.92 pp              |
| di32_slow_mamba | 7.70%       | 15.82%      | +8.12 pp              |

The slower-Mamba rollout score increases more with di (+1.11 pp) than the joint score (+0.62 pp).

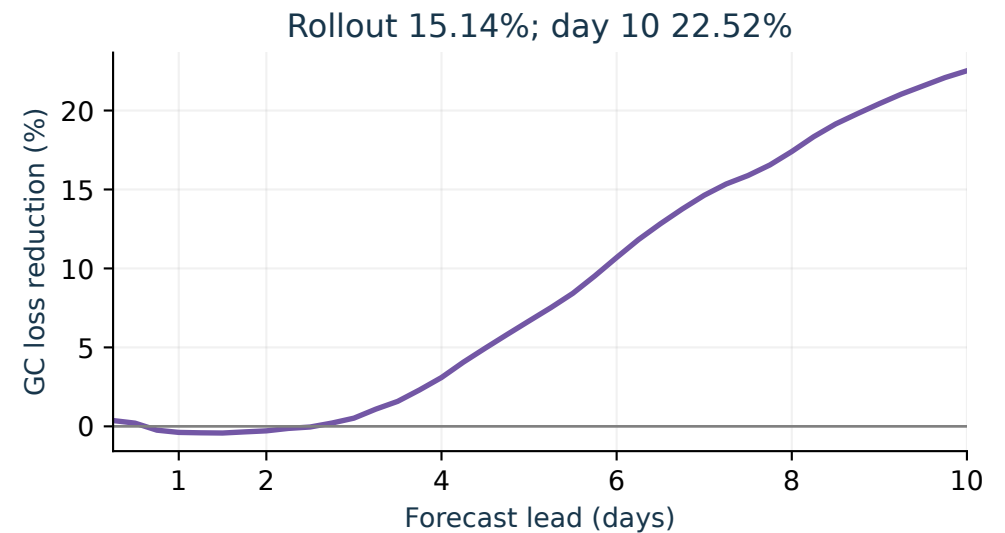
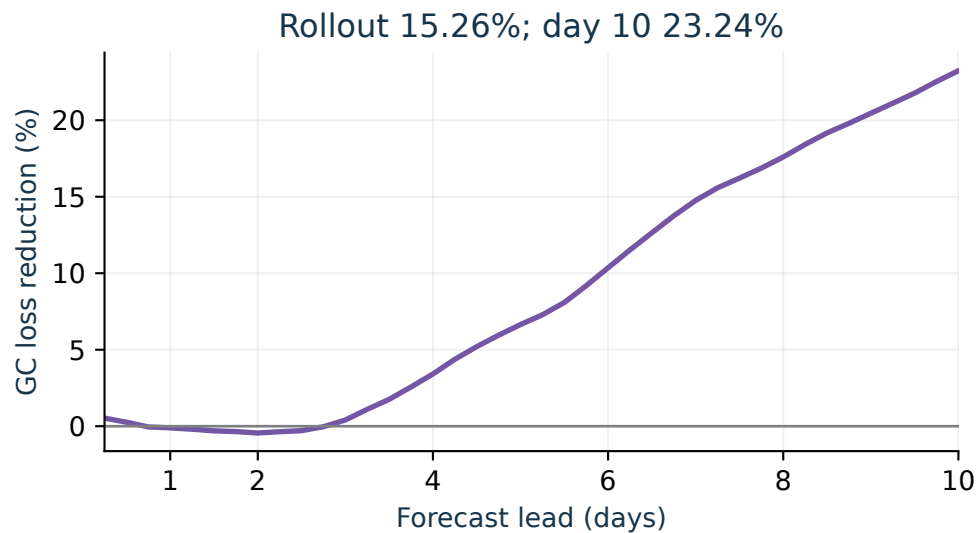
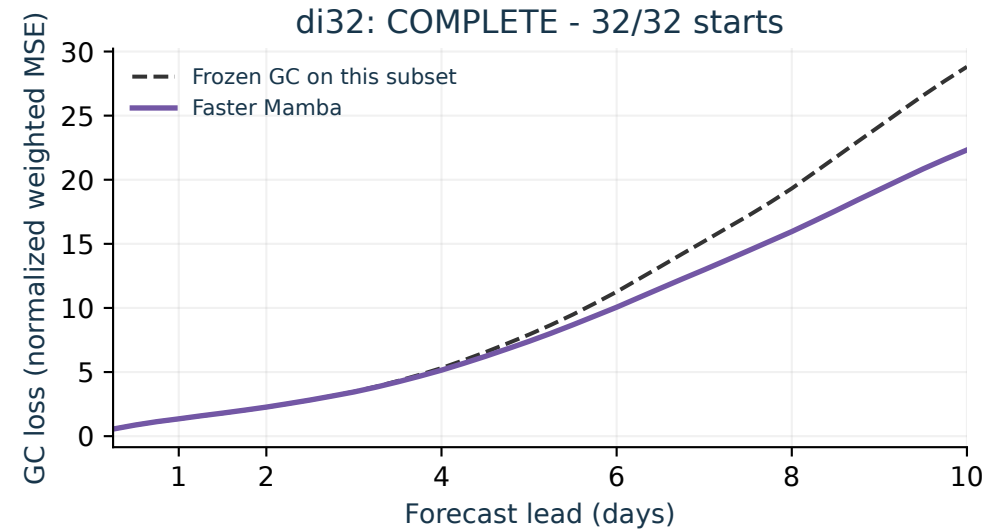
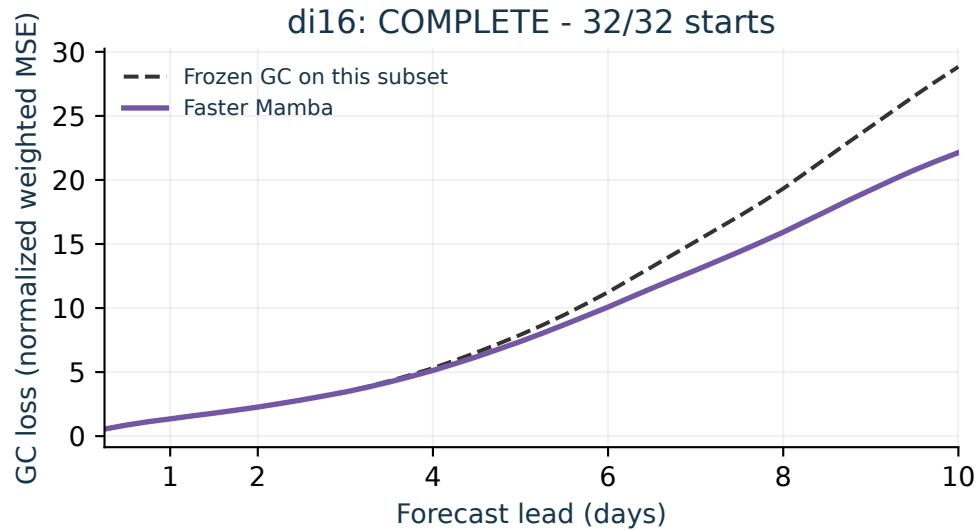
The gap to joint narrows from 0.97 pp at di16 to 0.48 pp at di32.

These are descriptive differences: all ten warm-up parameter hashes differ. In particular, di16 slower Mamba starts weaker but gains more during continuation. A shared-checkpoint comparison is needed to attribute the final-score difference to learning rate alone.

# Faster Mamba: evaluation progress

Peak Mamba LR 3e-4; spatial LR 1e-4 | final branch checkpoint shown here; SWA reported separately

**Both final checkpoints have completed the matched 32-start evaluation.**

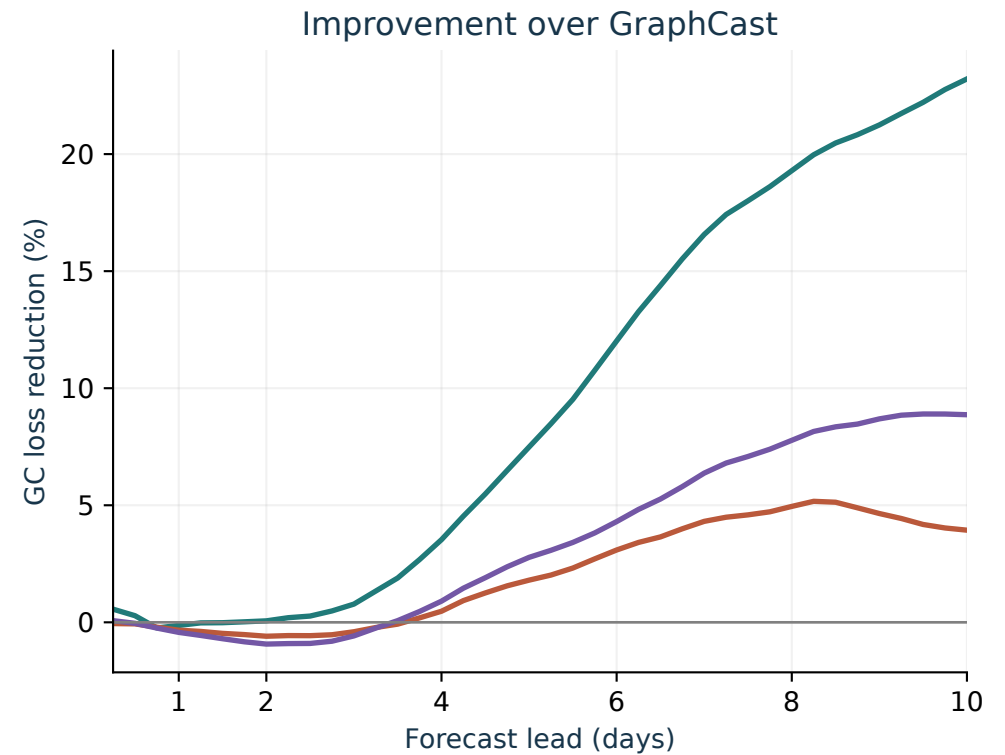
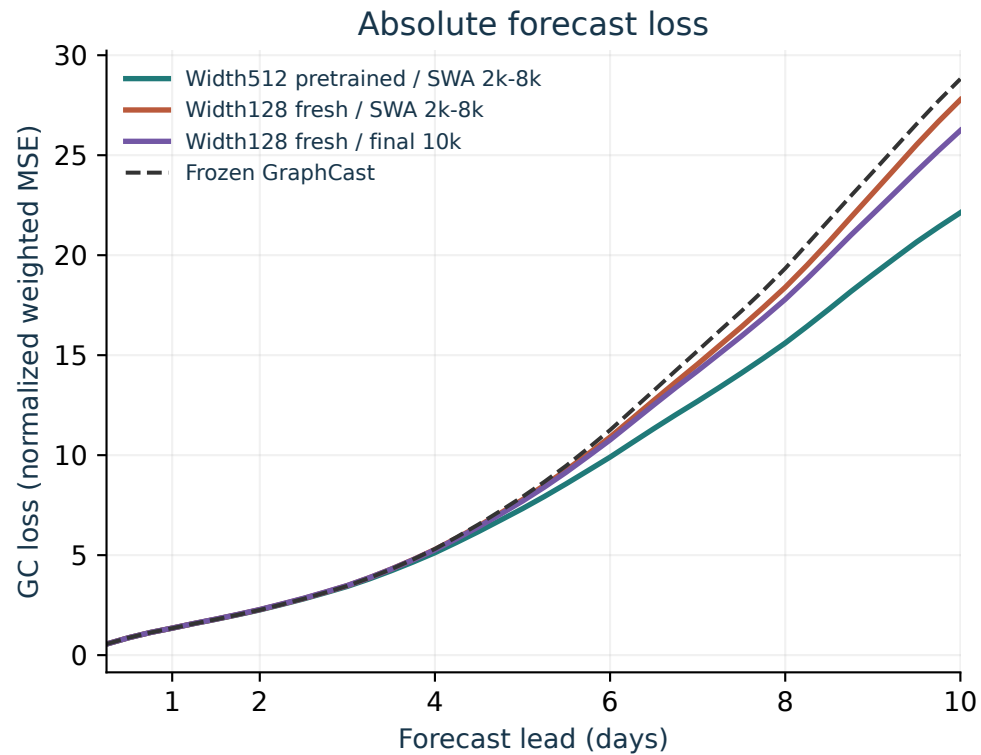


di16 SWA: complete 32/32 | di32 SWA: complete 32/32

Snapshots retain the exact input bytes and completed start IDs. Missing per-start losses prevent a matched-subset comparison.

# Smaller residual GraphCast: cost and skill

Separate batch-1 experiment | 10,000 updates | di16 / BCG1 | same 32 starts and 40 forecast leads



| Residual branch / artifact      | Parameters | Rollout reduction | Day-10 reduction |
|---------------------------------|------------|-------------------|------------------|
| Width512 pretrained / SWA 2k-8k | 10,348,879 | 16.28%            | 23.21%           |
| Width128 fresh / SWA 2k-8k      | 730,447    | 3.63%             | 3.94%            |
| Width128 fresh / final 10k      | 730,447    | 6.33%             | 8.87%            |

Width128 has 92.94% fewer residual parameters, retaining 22.28% of the reference SWA benefit.

Width and pretrained-versus-fresh initialization both change; this does not isolate width. Baseline width stays 512.

# All arms, methods and evidence

Completed-only score table | exact normalized, latitude-, pressure- and variable-weighted GraphCast MSE

| Arm                 | Warm-up rollout | Final rollout | SWA rollout | SWA day 10 |
|---------------------|-----------------|---------------|-------------|------------|
| di16_joint          | 9.06%           | 15.26%        | 15.67%      | 22.44%     |
| di16_slow_mamba     | 7.13%           | 13.63%        | 14.70%      | 21.98%     |
| di16_frozen_spatial | 9.24%           | 12.15%        | 11.59%      | 17.34%     |
| di32_joint          | 7.37%           | 15.69%        | 16.29%      | 23.32%     |
| di32_slow_mamba     | 7.70%           | 15.21%        | 15.82%      | 22.96%     |
| di32_frozen_spatial | 7.32%           | 12.02%        | 11.88%      | 17.85%     |
| di16_fast_mamba     | 9.45%           | 15.26%        | 16.34%      | 23.68%     |
| di16_slow_spatial   | 9.03%           | 15.71%        | 15.23%      | 22.08%     |
| di32_fast_mamba     | 7.94%           | 15.14%        | 15.73%      | 22.75%     |
| di32_slow_spatial   | 6.24%           | 15.83%        | 15.02%      | 21.99%     |

## METHOD

Loss reduction =  $100 \times (1 - \text{model loss} / \text{matched frozen-baseline loss})$ . Rollout losses aggregate all starts and 40 six-hour leads before the ratio; percentages are never averaged across leads. Training: 500 private warm-up + 2,000 branch updates; batch 4; seed 22; Mamba1; residual width512/MP2; di16 or di32; BCG1; state size16; 2 temporal layers. Peak LR 1e-4 with Mamba factors 0.3 / 1 / 3, 50-update warm-up, cosine to 1e-5 before group scaling; AdamW (0.9, 0.98), weight decay 1e-4. Evaluation: 2022, same 32 start IDs, cold full feedback, zero state, no history warm start. Checked completeness, IDs, metadata, loss ratios and arrays. Maximum matched baseline deviation: 0.556%. Partial previews have fewer starts; no common per-start losses are saved for retrospective matching.

## EVIDENCE AND REPRODUCTION

Report inputs, hashes and checks: v24\_mamba\_lr\_di\_10day\_2026-09-14.sources.json  
Numerical summary: v24\_mamba\_lr\_di\_10day\_2026-09-14.csv  
Input snapshots and per-lead CSV: plots/analyze\_models/data/resolution\_eval/v24\_mamba\_lr\_di\_10day\_2026-09-14/  
Generator: scripts/analyze\_models/report\_v24\_mamba\_lr\_di.py (GraphCast environment).  
One seed, differing warm-up weights and a development evaluation set: no significance claim.